

APPENDIX L — EXEMPLAR CLAIM-LIKE STATEMENTS (OPTIONAL, NON-LIMITING)

A. Overview

This appendix provides optional, non-limiting claim-like statements intended to help describe the invention in a structured way. A provisional application does not require claims. These statements are illustrative only and do not limit the disclosure.

B. Exemplary System Statement (Non-Limiting)

1. A system comprising: a) a plurality of sensing nodes including at least one intimate-region sensing node and at least one context node; and b) a computing system configured to receive data from the plurality of sensing nodes and to generate privacy-preserving outputs; wherein the computing system performs artifact rejection and validity gating using at least one quality index, derives one or more privacy-preserving metrics, and generates at least one correlation output. 2. The system of statement 1, wherein the plurality of sensing nodes includes at least one external wearable node configured for longitudinal measurement during sleep and/or daily life. 3. The system of statement 1, wherein the plurality of sensing nodes includes at least one intraluminal wearable node configured for longitudinal tissue-adjacent measurement. 4. The system of statement 1, wherein the plurality of sensing nodes includes at least one session-based geometry node configured to generate a baseline reference used to normalize longitudinal metrics. 5. The system of statement 1, wherein the plurality of sensing nodes includes at least one intraluminal geometry/compliance node configured to generate a stimulus-response profile and a compliance index. 6. The system of statement 1, wherein the context node includes one or more of ECG sensing, PPG sensing, IMU sensing, temperature sensing, and electrodermal activity sensing. 7. The system of statement 1, wherein the privacy-preserving outputs comprise one or more of indices, zones, ranges, curves, distributions, and confidence indicators without explicit anatomical imagery and without explicit anatomical labeling. 8. The system of statement 1, wherein the correlation output comprises cross-device correlation between derived metrics of different nodes. 9. The system of statement 1, wherein the correlation output comprises cross-user correlation under consent gating using derived metrics without sharing raw signals between users.

C. Exemplary Method Statement (Non-Limiting)

10. A method comprising: a) acquiring time-series data from at least one intimate-region sensing node; b) acquiring context data from at least one context node; c) computing at least one quality index and selecting valid windows; d) deriving privacy-preserving metrics from the valid windows; and e) presenting outputs as abstract indices, zones, ranges, curves, and confidence indicators. 11. The method of statement 10, further comprising generating a baseline reference using a geometry workflow and normalizing longitudinal metrics using the baseline reference. 12. The method of statement 10, further comprising computing cross-device correlation outputs including an alignment score and a synchrony curve using time-aligned windows. 13. The method of statement 10, further comprising performing cross-user correlation under consent gating using derived metrics and providing consent status indicators and share-scope indicators.

D. Exemplary Data Integrity and Validation Statements (Non-Limiting)

14. The system or method of any prior statement, wherein validity gating uses at least two of: motion gating, contact integrity checks, thermal plausibility checks, and multi-sensor agreement checks. 15. The system or method of any prior statement, wherein the computing system outputs a reduced-confidence indicator when validity thresholds are not met.

E. Variations

Any statement herein may be modified to include additional nodes, fewer nodes, substituted sensor modalities, different processing distributions (on-node, on-device, cloud), and different privacy modes (local-only, derived-only exchange, consent-gated exchange).